

AAHIL ALI

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EDUCATION

Cornell University

Expected May 2027

Major: Bachelor of Science in Mechanical Engineering, Dyson Business Minor for Engineers

Relevant Courses (*Currently Enrolled): Mechatronics*, Heat Transfer*, Propulsion of Aircrafts & Rockets*, Automotive Engineering*, Mechanics of Engineering Materials, System Dynamics, Fluid Dynamics, Thermodynamics, Dynamics

WORK EXPERIENCE

General Dynamics Electric Boat | Design & Engineering Co-op

Aug. 2026 – Present | Groton, CT

- Support engineering and procurement execution for COLUMBIA-class SSBNs, working directly with vendors to resolve fabrication and manufacturability challenges alongside conformance to procurement specifications and PO requirements.
- Review and disposition Vendor Information Requests by validating supplier technical data against engineering drawings, governing welding specifications, and 3D design models in Teamcenter.

Cornell Electric Vehicles Project Team | Steering Subteam

Oct. 2023 - Present | Ithaca, NY

- Collaborate with a group of 75+ to design and build a hyper-efficient autonomous EV for the Shell Eco-marathon, coming in 5th on the track and winning two off-track awards and \$4500 from the Shell Eco-marathon Americas competition.
- Designed and manufactured a lightweight acceleration pedal and sensor mounts in Autodesk Inventor; analyzed static forces in MATLAB to achieve 25 lbs of pedal feedback force; integrated a linear potentiometer to record position data for testing.
- Led steering system layout for 2026 vehicle and designed a threaded aluminum mounting plate that defines the interface for the rack and pinion, pedal box, and autonomy hardware, engineered for maximum stability and minimal weight; machined 50+ custom components using CNC, manual mill, and lathe to support vehicle assembly.

Lefiell Manufacturing | Manufacturing Engineering Intern

Summer 2026 | Santa Fe Springs, CA

- Developed Autodesk Inventor 3D models and process drawings for five Boeing main landing gear tubular components, translating customer drawings into machining operation sequences, intermediate part geometry, datum schemes, defining fixturing and work holding requirements, and applying GD&T per ASME Y14.5; all five parts released to production.
- Designed 10 custom thread plugs for RH/LH UNJF-3B threads per AS8879; created Autodesk Inventor assemblies for fit checks to verify thread engagement and drilling clearance, and iterated tooling geometry with CNC programmers.
- Created a gundrill depth gauge for dimensional verification of internal radius tangent point location on three tubular parts, using axially spaced gauging ridges matched to varying bore depths and consolidating three inspection gauges into one.
- Reverse engineered a main landing gear door forward stop fitting from a physical component and legacy Boeing hand drawing, using caliper measurements to reconstruct part geometry and model it in Autodesk Inventor for manufacturing.
- Validated sonobuoy launch components by cross checking CATIA V5 customer models against Autodesk Inventor models and engineering drawings, verifying dimensional accuracy, hole locations, and feature geometry before production.

Information & Decision Science Lab | Undergraduate Researcher

Aug 2025 – May 2026 | Ithaca, NY

- Contributing to the MathWorks-sponsored IDS³C Scaled Smart-City Testbed Project, a 1:25-scale research platform to test and validate autonomous vehicle control strategies in urban traffic systems.
- Implemented Arduino/Raspberry Pi hardware across 45 RC vehicles to enable scalable fleet testing and shorten debug cycles by 40%; created an engineering manual covering hardware/software integration & chassis updates, cutting onboarding time.

Cornell University MAE 2250 Mechanical Design | Teaching Assistant

Jan 2026 – May 2026 | Ithaca, NY

- Taught a weekly laboratory class of 25–35 undergraduate students on the topics of Computer-Aided Design, Machining, 3D Printing, Laser Cutting, Design for Manufacturing, Rapid Prototyping, plastic and metal part design, and other mechanical synthesis skills.

HydroClip

Jan 2025 – May 2025 | Ithaca, NY

- Created a backpack strap mounted water bottle holder; designed a sliding lock and latch geometry in Fusion 360, set clearances and tolerances for a secure fit, and iterated design with six 3D printed prototypes
- Conducted bench testing with pull, shake, and walk tests to adjust snap fit assembly and durability; refined latch geometry and strap fit to maximize strength in the direction of bending.

SKILLS

Technical: CAD (Inventor, Fusion, AutoCAD, SolidWorks), DFMA, GD&T, DFMEA, Master Modelling, Tolerance Stackup Analysis, ANSYS FEA (Mechanical, Fluent CFD), MATLAB, Arduino, Python, Excel, Office, PLEX MES, Teamcenter PLM

Manufacturing: Machining (Mill, Lathe, CNC), 3D Printing, Carbon Fiber (Vacuum Infusion, Wet Layup, Forged), Hand Tools